

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCVPE506A - ADVANCED ENVIRONMENTAL ENGG.

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1. What is the primary function of a waste stabilization pond (WSP)?
a) To decompose organic waste by mechanical means
b) To use microorganisms and sunlight to treat wastewater
c) To chemically neutralize hazardous waste
d) To provide a habitat for aquatic life
2. Which of the following wastewater treatment stages is designed to remove large solids and floatables from the water?
a) Primary treatment
b) Secondary treatment
c) Tertiary treatment
d) Sludge treatment
3. What is typically the main source of contaminants in municipal wastewater?
a) Industrial discharge and commercial waste
b) Household waste, including human waste and organic materials
c) Agricultural runoff
d) Toxic chemicals from manufacturing processes
4. Which of the following pollutants is more likely to be found in industrial wastewater compared to municipal wastewater?
a) Faecal coliforms
b) Solvents and oils
c) Human waste
d) Domestic cleaning agents
5. The primary function of oil separation by flotation is to:
a) Remove large suspended particles through settling
b) Remove oil and grease from wastewater by making it float to the surface
c) Enhance the biodegradation of organic materials
d) Decrease the volume of sludge produced during treatment
6. The primary function of oil separation by flotation is to:
a) Remove large suspended particles through settling
b) Remove oil and grease from wastewater by making it float to the surface
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d) Decrease the volume of sludge produced during treatment.
7. Environmental sanitation refers to:
a) The process of cleaning public spaces
b) The control of environmental factors that can adversely affect human health
c) The management of solid waste in urban areas
d) The installation of sewage systems only.
8. Communicable diseases are diseases that:
a) Cannot be transmitted between individuals
b) Are caused only by genetic factors
c) Spread from person to person through various routes
d) Are always treatable with antibiotics.

- 9 Waterborne diseases are primarily communicated through:a) Airborne transmissionb) Contaminated water sourcec) Direct person-to-person contactd) Vector-borne transmission
 - 10 Malaria is primarily spread by:a) Airborne transmissionb) Direct contact with infected individualsc) Mosquito bitesd) Contaminated food and water
 - 11 Which of the following diseases can be transmitted by the bite of an infected rodent?a) Typhoid b) Plague c) Hepatitisd) Influenza
 - 12 Which of the following is typically NOT a part of rural sanitation schemes?a) Constructing community toilets and household latrinesb) Promoting the use of composting toiletsc) Addressing waste treatment and managementd) Encouraging open defecation practices
- Q2. Solve the following.
- A) Explain with neat diagram the principle of waste stabilization pond 6
 - B) Describe the sources of industrial wastewater pollution 6
- Q3. Solve the following.
- A) Differentiate between industrial and municipal wastewater 6
 - B) What is self purification of natural streams. Also explain various actions involved in self purification 6
- Q4. Solve Any Two of the following.
- A) What do you mean by communicable diseases? Explain in detail 6
 - B) List out various low cost excreta disposal methods. Explain Aqua Privy and Borehole Privy 6
 - C) Explain rural sanitation in details 6
- Q5. Solve Any Two of the following.
- A) What do you mean by CETP? Explain the need of CETP equalization process in industrial wastewater treatment 6
 - B) Explain membrane filtration process with neat and clean diagram 6
 - C) Explain Life cycle of Mosquito 6
- Q6. Solve Any Two of the following.
- A) Explain following diseases with respect to infectious agent, occurrence, (i) Cholera (ii) Tuberculosis 6
 - B) Identify the sources of stream pollution 6
 - C) Describe oxidation ditch. Add a note on its advantages and disadvantages 6

*** End ***